RetailMax — Global Sales Analytics: Data Dictionary

What each of the 8 tables holds, every column with its type and sample values, and how the tables join — everything needed to write accurate SQL. No raw data is included.

#	Table	Size	What it holds	Primary Key
1	**Sales_Transactions**	50,000 rows × 21 columns	Every sales order line. One row = one product line within an order (Order_ID repeats when an order has multiple items).	Order_ID
2	**Customers_Master**	2,000 rows × 18 columns	Customer registry — one row per customer with demographics, segment and lifetime stats.	Customer_ID
3	**Products_Master**	500 rows × 16 columns	Product catalogue — one row per product with pricing, stock, supplier and rating info.	Product_ID
4	**Customer_Feedback**	5,000 rows × 10 columns	Product reviews — one row per feedback entry, linked to a customer, product and order.	Feedback_ID
5	**Employees**	50 rows × 13 columns	Employee registry incl. sales reps (REP-*) with department, salary, commission and manager.	Employee_ID
6	**Inventory_Log**	10,000 rows × 10 columns	Stock movements in/out of warehouses — one row per movement event.	Inventory_ID
7	**Marketing_Campaigns**	50 rows × 15 columns	Marketing campaigns with budget, funnel metrics (impressions→clicks→conversions) and revenue.	Campaign_ID
8	**Suppliers**	100 rows × 13 columns	Supplier directory with contact, payment terms, lead time and contract dates.	Supplier_ID

1. Sales_Transactions

50,000 rows × 21 columns | Every sales order line. One row = one product line within an order (Order_ID repeats when an order has multiple items). | Primary key: **Order_ID**

#	Column	Type	Example	2nd Example	Details
1	**Order_ID**	Text	ORD-500001	ORD-500002	Free text / ID — 50000 unique values | PK **[KEY]**
2	**Order_Date**	Text	2024-05-09	2025-03-09	Date (YYYY-MM-DD), range 2023-01-01 to 2025-06-19
3	**Order_Time**	Text	21:51	02:03	Free text / ID — 1440 unique values
4	**Customer_ID**	Text	CUST-11907	CUST-11362	Free text / ID — 2000 unique values | FK → Customers_Master **[KEY]**
5	**Product_ID**	Text	PRD-1483	PRD-1176	Free text / ID — 500 unique values | FK → Products_Master **[KEY]**
6	**Quantity**	Whole Number	1	1	Whole number, range 1 to 50
7	**Unit_Price**	Decimal Number	1569.15	804.77	Decimal number, range 9.31 to 2197.35
8	**Subtotal**	Decimal Number	1569.15	804.77	Decimal number, range 9.32 to 105696
9	**Discount_Percent**	Whole Number	10	5	Whole number, range 0 to 50
10	**Discount_Amount**	Decimal Number	156.92	40.24	Decimal number, range 0 to 47516
11	**Tax_Rate**	Decimal Number	0.19	0.2	Decimal number, range 0.08 to 0.2
12	**Tax_Amount**	Decimal Number	268.32	152.91	Decimal number, range 0.56 to 20101.9
13	**Shipping_Cost**	Decimal Number	47.27	26.38	Decimal number, range 0 to 50
14	**Total_Amount**	Decimal Number	1727.82	943.82	Decimal number, range 7.42 to 120611
15	**Payment_Method**	Text	PayPal	Cash on Delivery	Categorical — 7 values: Bank Transfer, Buy Now Pay Later, Cash on Delivery, Credit Card, Debit Card, Digital Wallet, PayPal
16	**Order_Status**	Text	Delivered	Delivered	Categorical — 6 values: Cancelled, Delivered, Processing, Refunded, Returned, Shipped
17	**Shipping_Method**	Text	Standard	Same Day	Categorical — 5 values: Express, Free Shipping, International, Same Day, Standard
18	**Delivery_Date**	Text	2024-05-23	2025-03-13	Date (YYYY-MM-DD), range 2023-01-03 to 2025-07-03
19	**Sales_Rep_ID**	Text	REP-48	REP-36	Free text / ID — 50 unique values | FK → Employees **[KEY]**
20	**Channel**	Text	Phone Order	Social Media	Categorical — 5 values: Marketplace, Mobile App, Online Store, Phone Order, Social Media
21	**Coupon_Code**	Text	WELCOME5	VIP25	Categorical — 5 values: FLASH20, SAVE10, SUMMER15, VIP25, WELCOME5

2. Customers_Master

2,000 rows × 18 columns | Customer registry — one row per customer with demographics, segment and lifetime stats. | Primary key: **Customer_ID**

#	Column	Type	Example	2nd Example	Details
1	**Customer_ID**	Text	CUST-10001	CUST-10002	Free text / ID — 2000 unique values | PK **[KEY]**
2	**First_Name**	Text	Dennis	Paul	Free text / ID — 100 unique values
3	**Last_Name**	Text	Adams	Torres	Free text / ID — 50 unique values
4	**Email**	Text	customer10001@email.com	customer10002@email.com	Free text / ID — 2000 unique values
5	**Phone**	Text	+20-223-5891	+77-533-2478	Free text / ID — 2000 unique values
6	**Country**	Text	Australia	France	Categorical — 10 values: Australia, Brazil, Canada, France, Germany, India, Japan, Mexico, UK, USA
7	**City**	Text	Brisbane	Bordeaux	Free text / ID — 99 unique values
8	**Address**	Text	8043 Maple Dr	6923 Park Rd	Free text / ID — 1951 unique values
9	**Postal_Code**	Whole Number	50645	90460	Whole number, range 10,032 to 99,998
10	**Date_of_Birth**	Text	2014-02-21	1976-09-19	Date (YYYY-MM-DD), range 1960-01-02 to 2019-12-06
11	**Registration_Date**	Text	2023-03-07	2022-02-14	Date (YYYY-MM-DD), range 2021-01-02 to 2024-12-31
12	**Customer_Segment**	Text	VIP	Regular	Categorical — 5 values: Enterprise, Inactive, New, Regular, VIP
13	**Loyalty_Points**	Whole Number	23673	6858	Whole number, range 0 to 99,869
14	**Avg_Order_Value**	Whole Number	1525	293	Whole number, range 10 to 4,989
15	**Total_Orders**	Whole Number	33	111	Whole number, range 0 to 200
16	**Preferred_Category**	Text	Books & Media	Toys & Games	Categorical — 10 values: Automotive, Beauty & Health, Books & Media, Electronics, Fashion, Groceries, Home & Kitchen, Office Supplies, Sports & Outdoors, Toys & Games
17	**Newsletter_Subscribed**	Text	Yes	Yes	Categorical — 2 values: No, Yes
18	**Account_Status**	Text	Active	Active	Categorical — 2 values: Active, Inactive

3. Products_Master

500 rows × 16 columns | Product catalogue — one row per product with pricing, stock, supplier and rating info. | Primary key: **Product_ID**

#	Column	Type	Example	2nd Example	Details
1	**Product_ID**	Text	PRD-1001	PRD-1002	Free text / ID — 500 unique values | PK **[KEY]**
2	**Product_Name**	Text	ActiveLife Professional Men Clothing Model 29	PlayWorld Ergonomic Accessories Pack 29	Free text / ID — 500 unique values
3	**Category**	Text	Fashion	Electronics	Categorical — 10 values: Automotive, Beauty & Health, Books & Media, Electronics, Fashion, Groceries, Home & Kitchen, Office Supplies, Sports & Outdoors, Toys & Games
4	**Subcategory**	Text	Men Clothing	Accessories	Free text / ID — 96 unique values
5	**Brand**	Text	ActiveLife	PlayWorld	Free text / ID — 30 unique values
6	**Base_Price**	Decimal Number	283.38	901.17	Decimal number, range 10.34 to 1999.15
7	**Cost_Price**	Decimal Number	122.07	435.68	Decimal number, range 5.53 to 1290.68
8	**Stock_Level**	Whole Number	4467	53	Whole number, range 1 to 4,988
9	**Reorder_Point**	Whole Number	94	438	Whole number, range 50 to 500
10	**Supplier_ID**	Text	SUP-175	SUP-120	Free text / ID — 100 unique values | FK → Suppliers **[KEY]**
11	**Rating**	Decimal Number	3.6	4.2	Decimal number, range 2.5 to 5
12	**Reviews_Count**	Whole Number	244	2787	Whole number, range 22 to 4,988
13	**Weight_kg**	Decimal Number	4.78	13.97	Decimal number, range 0.18 to 49.9
14	**Warranty_Months**	Whole Number	6	6	Whole number, range 0 to 36
15	**Launch_Date**	Text	2022-10-31	2025-05-14	Date (YYYY-MM-DD), range 2020-01-01 to 2025-06-14
16	**Is_Active**	Text	No	Yes	Categorical — 2 values: No, Yes

4. Customer_Feedback

5,000 rows × 10 columns | Product reviews — one row per feedback entry, linked to a customer, product and order. | Primary key: **Feedback_ID**

#	Column	Type	Example	2nd Example	Details
1	**Feedback_ID**	Text	FB-00001	FB-00002	Free text / ID — 5000 unique values | PK **[KEY]**
2	**Customer_ID**	Text	CUST-11455	CUST-10462	Free text / ID — 1828 unique values | FK → Customers_Master **[KEY]**
3	**Product_ID**	Text	PRD-1498	PRD-1137	Free text / ID — 500 unique values | FK → Products_Master **[KEY]**
4	**Order_ID**	Text	ORD-526850	ORD-500273	Free text / ID — 4761 unique values | FK → Sales_Transactions **[KEY]**
5	**Rating**	Whole Number	4	4	Whole number, range 1 to 5
6	**Review_Text**	Text	Delivery was fast but product is okay.	Amazing! Will buy again.	Categorical — 10 values: Amazing! Will buy again., Average quality, could be better., Best purchase ever!, Decent product for the price., Delivery was fast but product is okay., Exceeded my expectations!, Excellent product, highly recommended!, Good value for money., Not satisfied with the purchase., Poor quality, returning it.
7	**Review_Date**	Text	2024-09-02	2023-06-19	Date (YYYY-MM-DD), range 2023-01-01 to 2025-06-19
8	**Helpful_Count**	Whole Number	178	450	Whole number, range 0 to 500
9	**Verified_Purchase**	Text	No	Yes	Categorical — 2 values: No, Yes
10	**Would_Recommend**	Text	Yes	Yes	Categorical — 2 values: No, Yes

5. Employees

50 rows × 13 columns | Employee registry incl. sales reps (REP-*) with department, salary, commission and manager. | Primary key: **Employee_ID**

#	Column	Type	Example	2nd Example	Details
1	**Employee_ID**	Text	REP-10	REP-11	Free text / ID — 50 unique values | PK **[KEY]**
2	**First_Name**	Text	Alexander	Joseph	Free text / ID — 35 unique values
3	**Last_Name**	Text	Wilson	Rivera	Free text / ID — 36 unique values
4	**Email**	Text	rep10@retailmax.com	rep11@retailmax.com	Free text / ID — 50 unique values
5	**Department**	Text	Sales	Operations	Categorical — 5 values: Customer Support, IT, Marketing, Operations, Sales
6	**Job_Title**	Text	Sales Representative	Account Executive	Categorical — 5 values: Account Executive, Business Development, Sales Manager, Sales Representative, Senior Sales Rep
7	**Hire_Date**	Text	2019-10-14	2023-01-25	Date (YYYY-MM-DD), range 2018-01-25 to 2024-06-15
8	**Salary**	Whole Number	91243	59170	Whole number, range 36,343 to 117,113
9	**Commission_Rate**	Decimal Number	0.132	0.146	Decimal number, range 0.021 to 0.148
10	**Manager_ID**	Text	REP-17	REP-16	Categorical — 11 values: REP-10, REP-11, REP-12, REP-13, REP-14, REP-15, REP-16, REP-17, REP-18, REP-19, REP-20 | FK → Employees **[KEY]**
11	**Region**	Text	Europe	North America	Categorical — 4 values: Asia Pacific, Europe, Latin America, North America
12	**Performance_Rating**	Text	Excellent	Average	Categorical — 4 values: Average, Below Average, Excellent, Good
13	**Status**	Text	Active	On Leave	Categorical — 2 values: Active, On Leave

6. Inventory_Log

10,000 rows × 10 columns | Stock movements in/out of warehouses — one row per movement event. | Primary key: **Inventory_ID**

#	Column	Type	Example	2nd Example	Details
1	**Inventory_ID**	Text	INV-000001	INV-000002	Free text / ID — 10000 unique values | PK **[KEY]**
2	**Product_ID**	Text	PRD-1255	PRD-1114	Free text / ID — 500 unique values | FK → Products_Master **[KEY]**
3	**Movement_Type**	Text	Inbound	Outbound	Categorical — 5 values: Adjustment, Inbound, Outbound, Return, Transfer
4	**Quantity**	Whole Number	70	7	Whole number, range -50 to 1,000
5	**Unit_Cost**	Decimal Number	600.3	88.15	Decimal number, range 5.53 to 1290.68
6	**Total_Cost**	Decimal Number	42021.0	617.05	Decimal number, range -61426 to 1.22239e+06
7	**Warehouse_ID**	Text	WH-3	WH-2	Categorical — 10 values: WH-1, WH-10, WH-2, WH-3, WH-4, WH-5, WH-6, WH-7, WH-8, WH-9
8	**Date**	Text	2023-10-04	2024-08-02	Date (YYYY-MM-DD), range 2023-01-01 to 2025-06-19
9	**Reference_ID**	Text	REF-156774	REF-210550	Free text / ID — 9946 unique values
10	**Notes**	Text	Seasonal demand	Warehouse transfer	Categorical — 5 values: Customer return, Damaged goods, Regular restock, Seasonal demand, Warehouse transfer

7. Marketing_Campaigns

50 rows × 15 columns | Marketing campaigns with budget, funnel metrics (impressions→clicks→conversions) and revenue. | Primary key: **Campaign_ID**

#	Column	Type	Example	2nd Example	Details
1	**Campaign_ID**	Text	CAMP-001	CAMP-002	Free text / ID — 50 unique values | PK **[KEY]**
2	**Campaign_Name**	Text	Flash Sale 2023	Fall Launch 2025	Free text / ID — 44 unique values
3	**Campaign_Type**	Text	Display Ads	Influencer	Categorical — 8 values: Affiliate, Display Ads, Email, Influencer, Push Notification, SMS, Search Ads, Social Media
4	**Target_Category**	Text	Automotive	Sports & Outdoors	Categorical — 10 values: Automotive, Beauty & Health, Books & Media, Electronics, Fashion, Groceries, Home & Kitchen, Office Supplies, Sports & Outdoors, Toys & Games
5	**Start_Date**	Text	2024-11-15	2024-10-22	Date (YYYY-MM-DD), range 2023-01-08 to 2025-02-19
6	**End_Date**	Text	2025-01-15	2024-12-09	Date (YYYY-MM-DD), range 2023-02-10 to 2025-03-24
7	**Budget**	Whole Number	53491	74536	Whole number, range 1,218 to 98,723
8	**Actual_Spend**	Decimal Number	53014.82	62223.73	Decimal number, range 1132.16 to 88403.8
9	**Impressions**	Whole Number	9256185	7744083	Whole number, range 150,600 to 9,900,915
10	**Clicks**	Whole Number	772544	772690	Whole number, range 7,078 to 935,494
11	**Conversions**	Whole Number	87552	87544	Whole number, range 436 to 166,913
12	**Revenue_Generated**	Decimal Number	19746601.83	24732545.51	Decimal number, range 101534 to 4.30092e+07
13	**ROI_Percent**	Decimal Number	37147.32	39647.77	Decimal number, range 159.94 to 1.30019e+06
14	**Status**	Text	Completed	Completed	Categorical — 1 values: Completed
15	**Channel**	Text	Hybrid	Online	Categorical — 3 values: Hybrid, Offline, Online

8. Suppliers

100 rows × 13 columns | Supplier directory with contact, payment terms, lead time and contract dates. | Primary key: **Supplier_ID**

#	Column	Type	Example	2nd Example	Details
1	**Supplier_ID**	Text	SUP-100	SUP-101	Free text / ID — 100 unique values | PK **[KEY]**
2	**Supplier_Name**	Text	Global Supply Co 1	Global Supply Co 2	Free text / ID — 100 unique values
3	**Contact_Person**	Text	Katherine Brown	Emma Clark	Free text / ID — 100 unique values
4	**Email**	Text	contact100@supplier.com	contact101@supplier.com	Free text / ID — 100 unique values
5	**Phone**	Text	+13-206-6792	+22-500-4941	Free text / ID — 100 unique values
6	**Country**	Text	Canada	India	Categorical — 10 values: Australia, Brazil, Canada, France, Germany, India, Japan, Mexico, UK, USA
7	**City**	Text	Vancouver	Jaipur	Free text / ID — 60 unique values
8	**Payment_Terms**	Text	Cash on Delivery	Net 15	Categorical — 5 values: 2/10 Net 30, Cash on Delivery, Net 15, Net 30, Net 60
9	**Lead_Time_Days**	Whole Number	50	11	Whole number, range 3 to 60
10	**Rating**	Decimal Number	2.3	4.5	Decimal number, range 2 to 5
11	**Contract_Start**	Text	2019-11-30	2021-05-20	Free text / ID — 97 unique values
12	**Contract_End**	Text	2025-11-20	2027-01-01	Free text / ID — 95 unique values
13	**Status**	Text	Active	Active	Categorical — 2 values: Active, Inactive

How the Tables Join (verified against the data)

From Table	Join Column	To Table	Relationship
Sales_Transactions	Customer_ID	Customers_Master	Many customers → 1 customer
Sales_Transactions	Product_ID	Products_Master	Many sales → 1 product
Sales_Transactions	Sales_Rep_ID	Employees	Many sales → 1 sales rep
Customer_Feedback	Customer_ID	Customers_Master	Many reviews → 1 customer
Customer_Feedback	Product_ID	Products_Master	Many reviews → 1 product
Customer_Feedback	Order_ID	Sales_Transactions	Many reviews → 1 order
Inventory_Log	Product_ID	Products_Master	Many movements → 1 product
Products_Master	Supplier_ID	Suppliers	Many products → 1 supplier
Employees	Manager_ID	Employees	Self-join: employee → manager

Example SQL joins

1.
```sql
SELECT c.First_Name, c.Country, s.Total_Amount
FROM Sales_Transactions s
JOIN Customers_Master c ON s.Customer_ID = c.Customer_ID;
```

2.
```sql
SELECT p.Product_Name, p.Category, SUM(s.Quantity) AS Units_Sold
FROM Sales_Transactions s
JOIN Products_Master p ON s.Product_ID = p.Product_ID
GROUP BY p.Product_Name, p.Category;
```

3.
```sql
SELECT e.First_Name AS Rep, m.First_Name AS Manager
FROM Employees e
LEFT JOIN Employees m ON e.Manager_ID = m.Employee_ID;
```